# CARDIOTWIN SENTINEL v4.1 — END-TO-END MULTIMODAL DIGITAL TWIN ARCHITECTURE



Proactive Cardiovascular & Cardiometabolic Screening Pipeline | Team Root Access (Harsh Mishra · VNIT Nagpur) | Happiest Health Digital Twin Challenge 2026

## STREAM 1: STATIC & 7D EHR

### Clinical & Biochemical Record

- Demographics, Vitals, BMI
- Lipids: Total Chol, LDL, HDL, TG
- Glycemic & Inflam: HbA1c, hs-CRP
- Active Meds: Beta-Blocker, Statin

### 7-Day Wearable Trajectory

- 7d Resting HR & Nocturnal RMSSD
- 7d Sleep Efficiency (%) & Steps

### Framingham x 1.45 South Asian

- D'Agostino Cox PH 10-Yr Base Risk
- Biological Vascular Age Model

## STREAM 2: 100 Hz EDGE NODE

### Hardware (BOM: INR 1,165 / $14)

- ESP32-WROOM-32D (240 MHz)
- MAX30102 Optical PPG (880nm IR)
- SH1106 128x64 I2C OLED + LiPo

### FreeRTOS Dual-Core Split

- Core 1 (Prio 5): 10,000 us ISR Deterministic 100 Hz -> chunkQueue
- Core 0 (Prio 1): Async WiFi POST

### Telemetry Integrity Contract

- seq, device_boot_id, deltas_us[]
- Zero network stalls on Core 1

## STAGE 2: DECOUPLED GATE

### Why Decoupled Gating Matters

Standard filters reject irregular beats as noise—discarding AFib. CardioTwin separates physics from physiology:

### Axis 1: Physical Gate (Blocks AI)

1. Contact Floor: ADC < 20,000 -> Status: FINGER_OFF
2. Amp Ceiling: ADC > 260,000 or Clipping > 5% -> SATURATED
3. Timing Jitter: |fs - 100| > 15Hz or Drops > 5% -> POOR_TIMING
4. Morphology: < 3 systolic peaks in 10s (0.5-8Hz Butterworth)
5. Stability: Requires >= 3 consecutive clean 10s windows

### Axis 2: Physiology (Passes to AI)

- Extreme Rate (<40 or >180 BPM) -> Passes with RECHECK tag
- Chaotic Intervals (RR-CV > 0.35) -> Passes as IRREGULAR_RHYTHM

## STAGE 3: DUAL-MODALITY AI

### Training Cohort (Zero Synthetic)

- 2,271 Unique ICU/Amb Patients
- 4,683 Windows (10s @ 100 Hz)
- MIMIC-III + CinC 2015 + BUT + BIDMC
- Patient-Isolated GroupKFold (k=5)

### Modality A: Classical Ensemble

- 27 HRV, Spectral & Morph Features
- XGBoost + Random Forest + ET
- Standalone Macro-F1: 48.00%

### Modality B: Inception-1D CNN

- Raw 1,000-Sample 1D Waveform
- 4 Parallel Kernels (3, 7, 15, 31)
- 393,224 Params (1.61 MB, 4.2ms)
- Standalone Macro-F1: 51.81%

### Temperature-Scaled Late Fusion

- 0.38 Classical (Tc=0.85) + 0.62 Inception-1D (Tdl=1.15)
- 8-Class Macro-F1: 54.40%
- 3-Tier Triage Acc: 75.68%
- Normal Specificity: 98.83%
- Tier-2 Anomaly Recall: 90.09%

## STAGE 4: DIGITAL TWIN CORE

### 2-Min Seated Personal Baseline

- 30s settling + >=6 clean windows
- Non-parametric Median + MAD
- Robust Z-Score departure index
- 30-day TTL & device_id lock

### EHR Vulnerability Coupling (M_EHR)

- Couples Stream 1 EHR + Stream 2:
- Beta-Blockade Escape (BPM >= 90 despite active Metoprolol)
- Subclinical AFib Prodrome
- Autonomic Vagal Withdrawal
- M_EHR Multiplier: 1.00x - 1.40x

### 2h / 6h Proactive Forecaster

- Fuses 7-day nocturnal RMSSD slope + acute dI/dt velocity
- Predicts Time-to-Decompensation 2 to 6 hours before crisis

### Dual-Horizon What-If Simulator

- 24-Hr Hemodynamic Twin Response
- 10-Yr ASCVD Risk (SGLT2i, Statin, Beta-Blocker, ACE-I, SBP, Smoking)

## STAGE 5: CLINICIAN & FHIR

### Clinician Command Portal

- React 19 + Socket.IO Live UI
- React Native Mobile Companion
- 100 Hz SVG Oscilloscope
- Live HRV Strip (RMSSD, SDNN, RR-CV, Kurtosis SQI)
- 8-Biomarker EHR Lab Panel
- 7-Day Wearable Sparklines

### Explainability & Honesty

- SHAP TreeExplainer top drivers
- Machine-Readable Evidence Ledger (No fake SpO2, PPG!=ECG, 77.08% ICU source audit)

### ABDM & HL7 FHIR R4 Export

- GET /fhir/export Bundle
- 14-Digit ABHA ID Linked
- LOINC 8867-4, 80404-7, 79423-0
- e-Sanjeevani Tele-Consult Ready

### DPDP Act 2023 Edge Privacy

- Raw waveforms purged in RAM

| INR 1,165 (~$14) | 100 Hz / +-0.4ms | 2,271 Patients | 98.83% / 90.09% | 2h & 6h Horizon | 54 / 54 Passing |
|---|---|---|---|---|---|
| Complete Edge Hardware BOM | FreeRTOS Core 1 ISR Sampling | Zero-Leakage GroupKFold Cohort | Normal Spec / Tier-2 Anomaly Recall | Proactive Decompensation Forecast | Automated Unit & Target Validation Tests |

# CARDIOTWIN SENTINEL v4.1 — HARDWARE SCHEMATIC, MATHEMATICAL FORMULATION & FHIR R4 SPECIFICATION



Zero-Guesswork Engineering Reference | 17 REST Endpoints + WebSocket Telemetry | 54 Automated Verification Tests

## 1. HARDWARE PINOUT & BILL OF MATERIALS (INR 1,165)

### I2C Circuit Pinout (400 kHz Fast Mode + 4.7k Pull-ups)

- ESP32 3V3 / GND -----> MAX30102 VIN / GND & OLED VCC / GND
- ESP32 GPIO 21 (SDA) --> MAX30102 SDA & SH1106 OLED SDA
- ESP32 GPIO 22 (SCL) --> MAX30102 SCL & SH1106 OLED SCL
- TP4056 + 650mAh LiPo -> ESP32 VIN / GND (6-8h Portable Run)

### Sensor Configuration (MAX30102 Register Setup)

- Wavelength: 880nm Infrared LED @ 6.0 mA Drive Current
- ADC Resolution: 18-bit (262,144 counts), 411 us Pulse Width
- Sampling Rate: 100.0 Hz (10,000 us hard timer period)

### Component Cost Breakdown (Single Unit vs 1k Batch)

- ESP32-WROOM-32D Dual-Core MCU: INR 480 ($5.80)
- MAX30102 Optical PPG Subsystem: INR 220 ($2.65)
- SSD1306 / SH1106 0.96" I2C OLED: INR 140 ($1.70)
- TP4056 USB-C Charger + 650mAh LiPo: INR 210 ($2.55)
- Biocompatible 3D PLA Finger Clip + Foam: INR 80 ($0.95)
- 4.7k Pull-ups, Bypass Caps, Silicone: INR 35 ($0.42)

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**TOTAL UNIT BOM: INR 1,165 (~$14.07) | < INR 950 @ 1,000 units**

## 2. DIGITAL TWIN MATHEMATICAL FORMULATIONS

### A. Non-Parametric Personal Baseline & Robust Z-Score

- Empirical 2-Min Seated Calibration (6+ clean 10s windows):
  Robust_Z(x) = (x_current - Median) / (1.4826 * max(MAD, min_mad))
  (50% statistical breakdown point; immune to transient spikes)

### B. Multimodal Personal Instability Index (0 - 100)

- I_raw = 0.35*S_rate + 0.25*S_irreg + 0.20*S_vagal + 0.20*S_persist
- EMA Smoothing: I_t = 0.35 * I_raw + 0.65 * I_(t-1)

### C. EHR-Coupled Vulnerability Substrate Multiplier (M_EHR)

- M_EHR = 1.0 + 0.10*I(CAD/PCI) + 0.08*I(HbA1c>=7.0)
  + 0.08*I(hs-CRP>2.0) + 0.14*I(Beta-Blockade Escape)
- Beta-Blockade Escape fires when BPM >= 90 despite active
  Metoprolol / Bisoprolol therapy (sympathetic breakthrough).

### D. 2-Hour & 6-Hour Proactive Horizon State Extrapolation

- I_hat(t + dt) = clip(I_t + (b_acute*dI_dt + b_7d*Slope_7d) * M_EHR)
- Fuses 7-day nocturnal RMSSD decay slope with acute velocity.

### E. South Asian Recalibrated Framingham 10-Yr Cox PH Risk

- Base_Risk = 1 - S0(10)^exp(Sum(beta_k * ln(X_k)) - mu)
- Recalibrated_Risk = min(99.0%, Base_Risk * 1.45)

## 3. ABDM HL7 FHIR R4 & API CONTRACT

### HL7 FHIR R4 Resource Mapping (GET /fhir/export)

- Patient: 14-digit ABHA ID (healthid.abdm.gov.in)
- Device: cardiotwin-esp32-01 (Firmware v4.1.0)
- Observation (HR): LOINC 8867-4 (beats/min)
- Observation (RMSSD): LOINC 80404-7 (ms)
- Observation (Rhythm): SNOMED-CT 361137007
- Observation (Instability): CARDIOTWIN-INSTABILITY
- RiskAssessment (10Y CVD + 6H Forecast): LOINC 79423-0

### Primary Backend REST & WebSocket Endpoints

- POST /data & /ingest_telemetry (ESP32 50-sample chunks)
- GET /twin/status (Full fused EHR + Live + 2h/6h state)
- POST /twin/simulate (Dual-Horizon 24h + 10yr What-If)
- POST /twin/scenario (Inject AFib, Tachy, Motion, Liftoff)
- POST /baseline/calibrate & GET /baseline/status
- GET /signal/gate & GET /telemetry/integrity
- GET /explainability/shap & /transparency/evidence_ledger

### Automated Test Verification Suite (54 / 54 Passing)

- test_cardiotwin.py (40 unit & API contract tests)
- test_instability_target_validation.py (14 target tests)

*Clinical Regulatory Boundary: CardioTwin Sentinel is an investigational clinical decision-support prototype. Single-channel optical PPG is not a 12-lead ECG and requires confirmatory physician review.*